Where else might the bumblebees be?

A hands-on exercise in species distribution modelling

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The conservation question

Where might a bumblebee occur beyond the places where it has been recorded? In this exercise you will use observations and climate maps to build a **species distribution model (SDM)**, then decide what its predictions can—and cannot—tell a conservation manager.

By the end, you should be able to:

1. **Map** recorded occurrences and identify limitations of opportunistic observations.
2. **Build and interpret** a climate-based model of relative suitability.
3. **Evaluate** its predictions and explain what additional evidence conservation decisions need.

Your outputs: a map of observations, a map of predicted climatic suitability, response curves and a **short report** answering four interpretation questions (see the final section).

Choose a species

Each group selects one Swedish bumblebee. The same code works for all four; comparing the outputs will reveal how different species relate to Sweden's varied climates.

Code key	Swedish name	Scientific name	Records supplied
`"mosshumla"`	Mosshumla	*Bombus muscorum*	757
`"berghumla"`	Berghumla	*Bombus monticola*	442
`"blaklockshumla"`	Blåklockshumla	*Bombus soroeensis*	5,307
`"ljunghumla"`	Ljunghumla	*Bombus jonellus*	1,852

Before you begin

1. Download the SDM exercise repository as a ZIP (or clone it). Open `sdm-exercise.Rproj` in RStudio. The `data` directory should sit alongside this project file.
2. Install the required R packages **before class**, if possible. All observations, maps and climate data are supplied in the repository; no data downloads are needed during the exercise. Maxnet does not require Java.
3. Run the chunks **in order**. Change only `species_key` to switch species. Figures will be saved to `results/<species_key>/`. The complete plotting code remains available in this Rmd and in its HTML rendering, even where omitted from the printed PDF.

1. Get ready: load packages and choose a bee

Do: Run the setup chunk, then change the first line of the species chunk to select your bee.

```
packages <- c("SDMtune", "terra", "sf", "ggplot2", "plotROC")
missing <- packages[!vapply(packages, requireNamespace, logical(1),
                            quietly = TRUE)]
if (length(missing)) {
  stop("Install these R packages before class: ", paste(missing, collapse = ", "))
}
# Run once beforehand if necessary:
# install.packages(c("SDMtune", "terra", "sf", "ggplot2", "plotROC"))
library(SDMtune)
library(terra)
library(sf)
library(ggplot2)
library(plotROC)
set.seed(42)  # Repeatable thinning and background sampling
```

```
species_key <- "mosshumla"  # Or "berghumla", "blaklockshumla", "ljunghumla"
species <- list(
  mosshumla = list(file = "mosshumla.csv",
                   common = "Mosshumla", latin = "Bombus muscorum"),
  berghumla = list(file = "berghumla.csv",
                   common = "Berghumla", latin = "Bombus monticola"),
  blaklockshumla = list(file = "blåklockshumla.csv",
                        common = "Blåklockshumla", latin = "Bombus soroeensis"),
  ljunghumla = list(file = "ljunghumla.csv",
                    common = "Ljunghumla", latin = "Bombus jonellus")
)
stopifnot(species_key %in% names(species))
bee <- species[[species_key]]
out_dir <- file.path("results", species_key)
dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)
bee$latin
```

```
## [1] "Bombus muscorum"
```

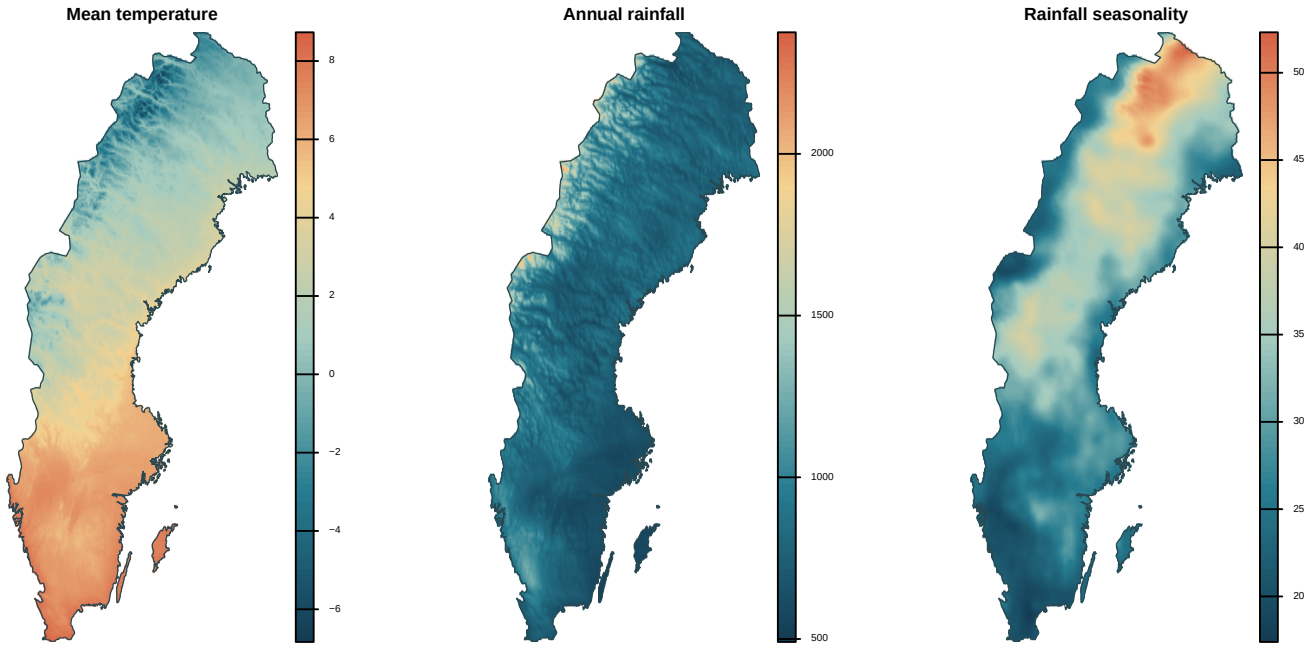
2. Explore the climate data

Why this matters. A climate-based SDM depends on the environmental variables we give it. We use just three CHELSA variables to make the patterns easier to explore: annual mean temperature (**BIO1**), annual precipitation (**BIO12**) and precipitation seasonality (**BIO15**). We use Sweden as the same study area for all four bees.

Do: Load the prepared layers, then compare their geographic patterns. The model will use linear and quadratic features to allow curved relationships.

```
inputs <- c("data/sweden.geojson", "data/climate_sweden.tif")
if (!all(file.exists(inputs))) {
  stop("Missing prepared files: ", paste(inputs[!file.exists(inputs)],
                                         collapse = ", "),
       ". Ask your teacher for the complete project download.")
}
sweden <- terra::vect("data/sweden.geojson")
climate <- terra::rast("data/climate_sweden.tif")
stopifnot(terra::nlyr(climate) == 3)
names(climate) <- c("bio1", "bio12", "bio15")
cols <- grDevices::colorRampPalette(
  c("#143D55", "#287D91", "#A5CCC0", "#F4D392", "#D76346")
)(100)
names(climate)
```

```
## [1] "bio1"  "bio12" "bio15"
```

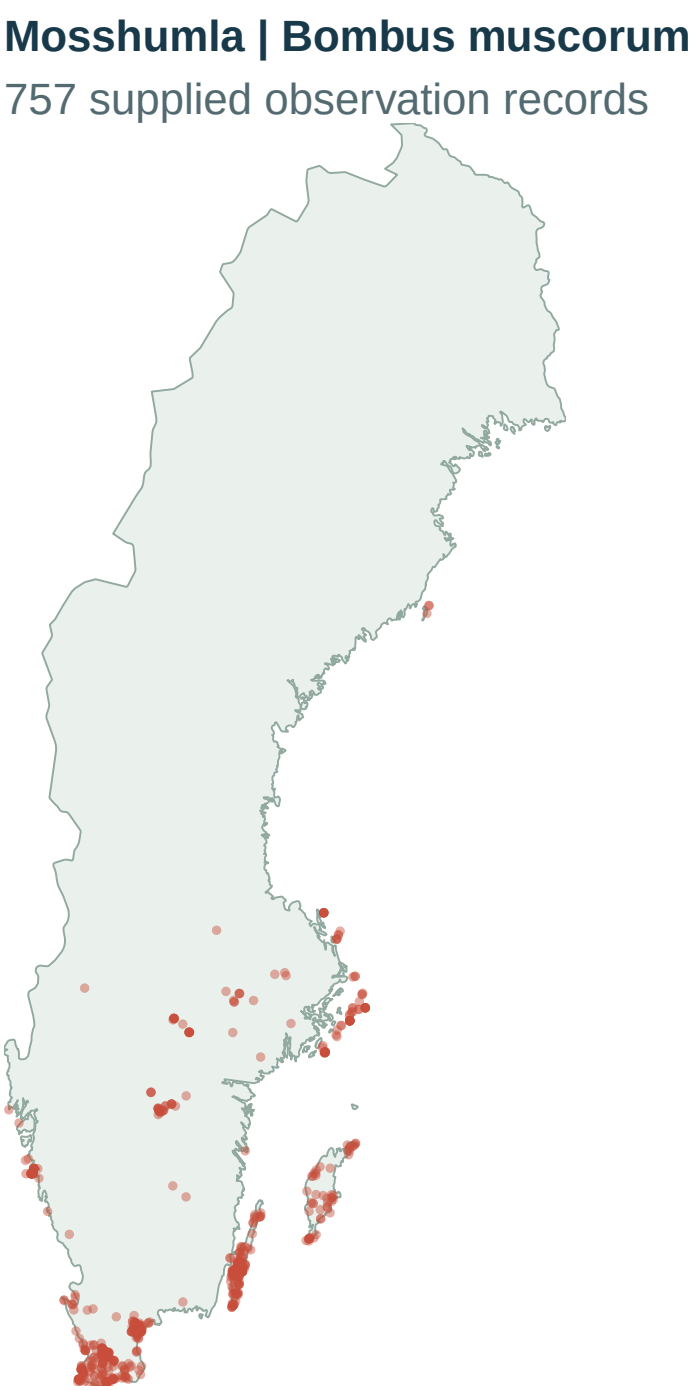


Pause & interpret. How do temperature and precipitation vary across Sweden? Apart from climate, what might make one region more suitable for your bee than another?

3. Map where the bee has been observed

Why this matters. The supplied data are observation records, **not** a systematic survey. Places without dots are not necessarily places without bees. Coordinates are supplied in the Swedish RT90 system (EPSG:3021), so we transform them before mapping.

```
records <- read.csv(file.path("data", bee$file), sep = ";",
                    fileEncoding = "UTF-8-BOM")
stopifnot(all(c("Ost", "Nord") %in% names(records)))
observations <- sf::st_as_sf(records, coords = c("Ost", "Nord"),
                             crs = 3021)
observations <- sf::st_transform(observations, 4326)
```



Pause & interpret. Do the dots show the bee's distribution, the places people are most likely to record it—or both? Name one location where additional surveys would help.

4. Fit a model: presence versus background

Why this matters. Maxnet compares the climates at **recorded locations** with climates available at **background locations** across Sweden. Background points are **not confirmed absences**. To avoid repeatedly counting records from the same climate grid cell, we retain at most one occurrence per cell. Records outside the climate grid remain on the observation map but cannot contribute to model fitting.

Do: Create presence and background samples, hold back 20% of presence cells for evaluation, and fit a simple Maxnet model.

```
xy <- as.data.frame(sf::st_coordinates(observations))
names(xy) <- c("x", "y")
presence <- as.data.frame(SDMtune::thinData(xy, env = climate))
background <- terra::spatSample(climate[[1]], size = 2000,
                                method = "random", na.rm = TRUE,
                                xy = TRUE, values = FALSE)
```

```r
cat("Original records:", nrow(records), "\n")
```

```
## Original records: 757
```

```r
cat("Cells retained for modelling:", nrow(presence), "\n")
```

```
## Cells retained for modelling: 245
```

```r
samples <- SDMtune::prepareSWD(species = bee$latin,
                               p = presence, a = background,
                               env = climate)
parts <- SDMtune::trainValTest(samples, test = 0.2,
                               only_presence = TRUE, seed = 42)
model <- SDMtune::train(method = "Maxnet", data = parts[[1]],
                        fc = "lq", reg = 1)
model
```

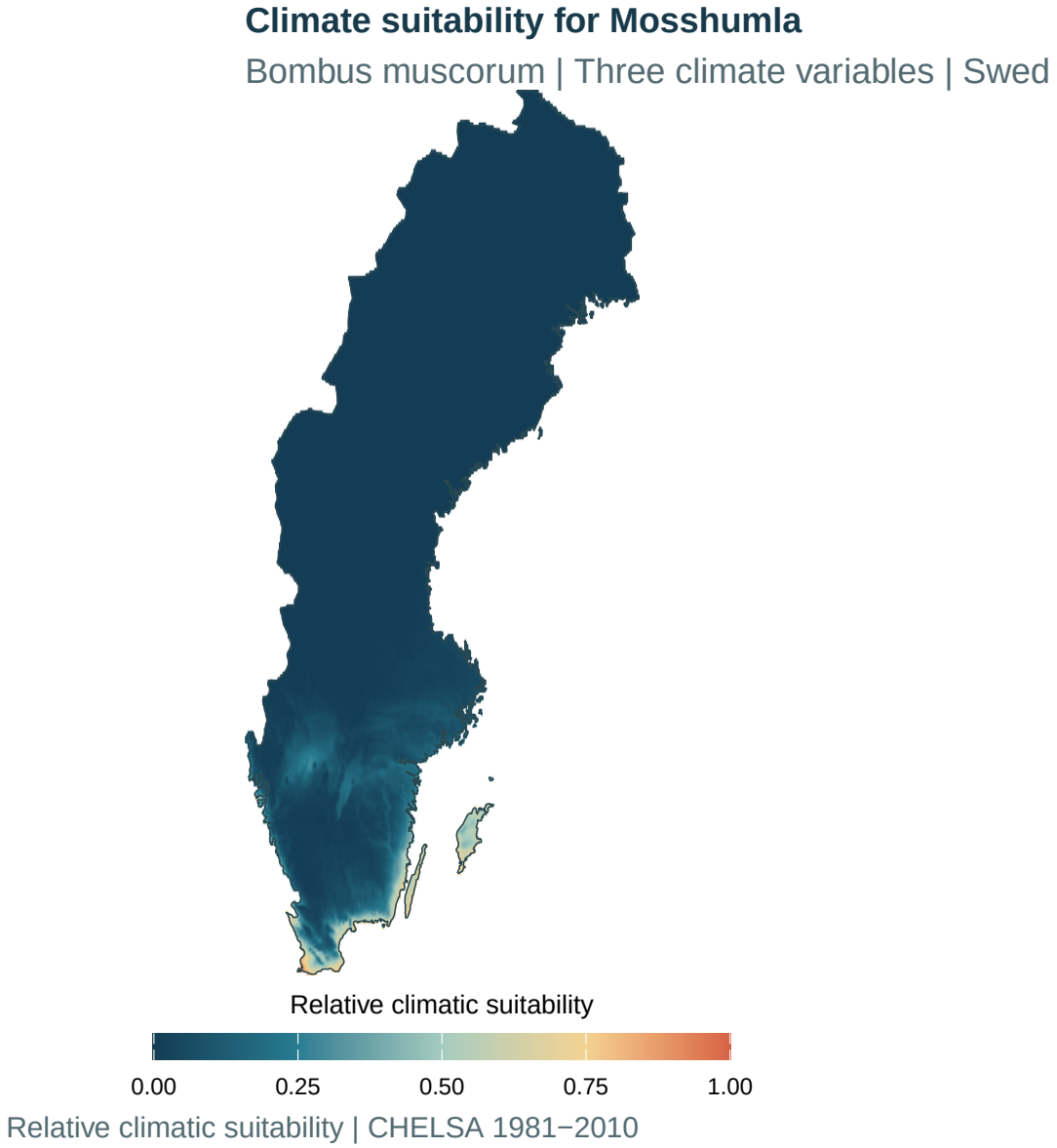
> **Pause & interpret.** Why do we remove duplicate climate-grid cells? How would the interpretation change if our background points were genuine surveyed absences?

5. Predict climate suitability across Sweden

Do: Use the fitted model to predict a score for every climate grid cell, then compare the prediction with your observation map.

Remember: This is a **relative climatic suitability score**, not a confirmed range boundary, a census or a calibrated probability of occupancy. Blue indicates lower modelled suitability and coral higher suitability.

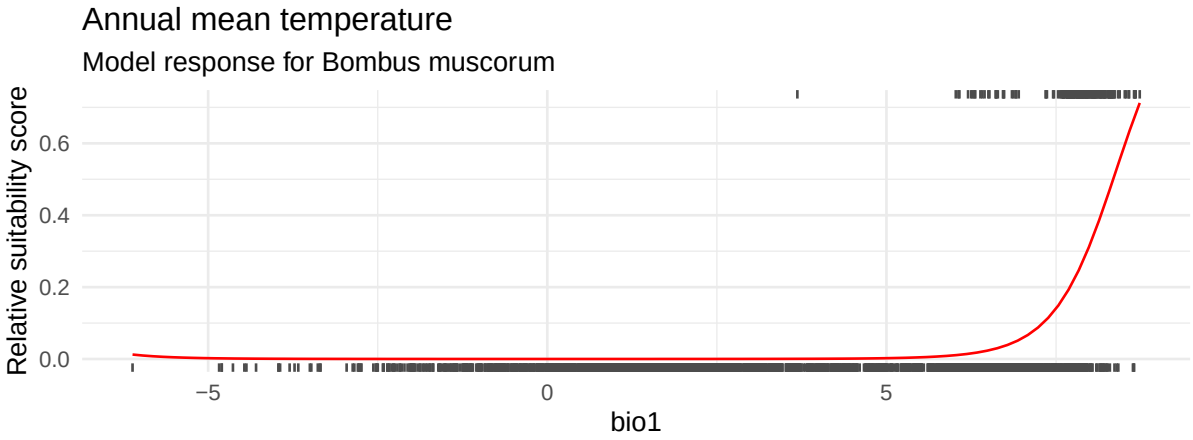
```r
suitability <- predict(model, data = climate, type = "logistic")
palette <- c("#143D55", "#287D91", "#A5CCC0", "#F4D392", "#D76346")
map_cells <- as.data.frame(suitability, xy = TRUE, na.rm = TRUE)
names(map_cells)[3] <- "score"
```

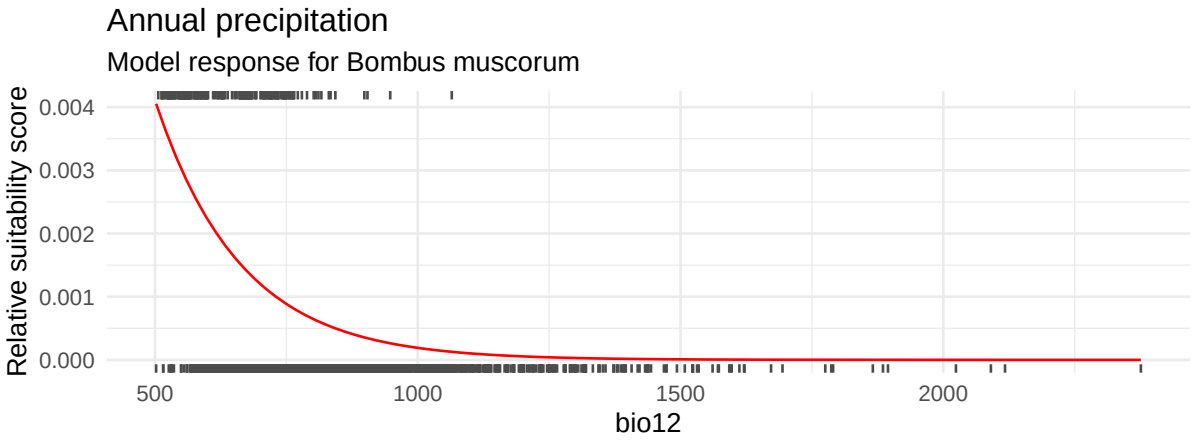


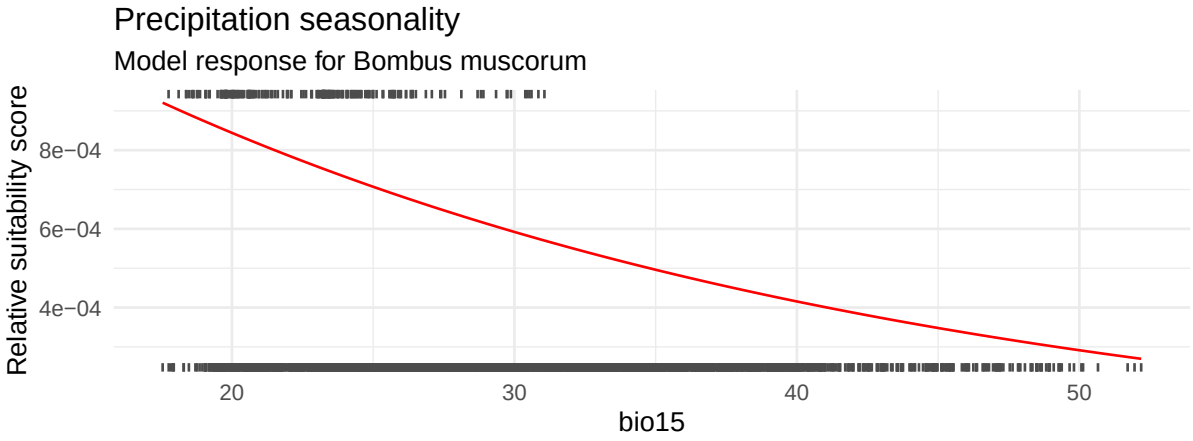
Pause & interpret. Where does the model predict suitable climate but there are few records? What might explain this gap? Are any records in areas with relatively low predicted suitability?

What relationships did the model learn?

Response curves vary **one predictor at a time** while holding the other two at their means. The tick marks show predictor values represented in the data. Interpret the shape of each curve cautiously, particularly at sparsely sampled values and where climate predictors are correlated.







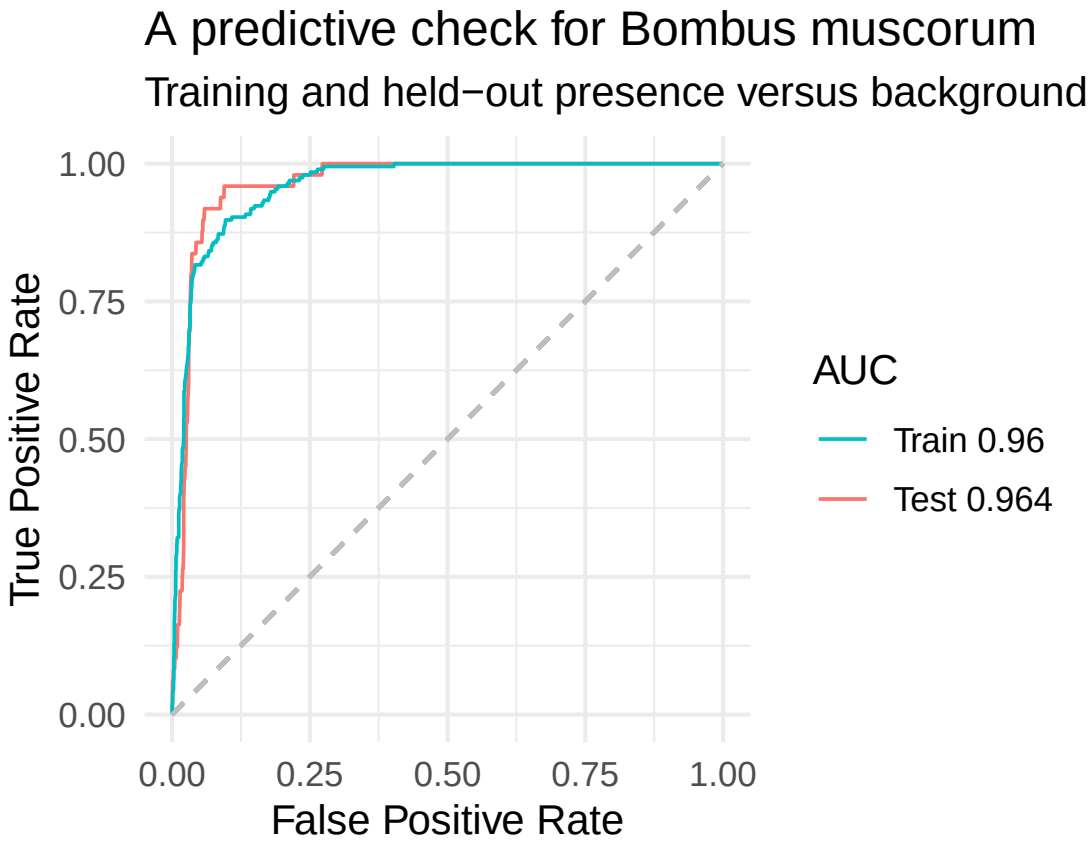
Pause & interpret. What is the *shape* of each relationship? Where is the evidence sparse? The y-axis scales differ: why is it risky to rank the importance of the predictors just by comparing these curves? Do the curves establish causation?

6. Evaluate the prediction

Why this matters. A model can produce an attractive map without making reliable predictions. We withheld 20% of presence cells. **AUC** measures how often the model ranks a withheld presence more highly than a background location; it does not tell us whether the predicted suitability is a true occupancy probability.

```
train_auc <- SDMtune::auc(model)
test_auc <- SDMtune::auc(model, test = parts[[2]])
cat(sprintf("Training AUC: %.3f | Held-out AUC: %.3f\n",
            train_auc, test_auc))
```

```
## Training AUC: 0.960 | Held-out AUC: 0.964
```



Interpret with care. Nearby records may share similar climates, even when randomly split between training and test sets. AUC also depends on the background area: distinguishing records from the whole of Sweden may be easier than predicting nearby localities. Neither AUC nor the map reveals whether flowers, nesting resources, land use, disease, dispersal or recorder effort limit the species. A fuller assessment would examine observation quality and dates, use spatial validation, and consider additional predictors. See SDMtune's evaluation guide.

> **Pause & interpret.** Is a high held-out AUC sufficient to recommend habitat protection at a particular location? What other information would a conservation manager need?

7. Your short conservation report

Use your own species and figures to write a brief, evidence-based interpretation. **Include** the Swedish and scientific species names, 2–3 sentences about its ecology (foraging, nesting or habitat), your **climate suitability map**, and references to your response curves and held-out AUC.

Answer these four questions:

1. **Where is the climate most suitable?** Describe the spatial pattern and point to relevant areas on your map.
2. **Which climatic conditions are associated with suitability?** Interpret the response curves and their limitations; do not assume they identify a physiological optimum.
3. **How persuasive is the model?** Explain what held-out AUC tells us and what it does *not* tell us.
4. **What would you advise a conservation manager?** Discuss whether climate alone is enough to assess habitat suitability and identify useful further evidence or surveys.

The central takeaway: A climate suitability map is a hypothesis about where a species *might* occur. Conservation decisions require attention to sampling, habitat, dispersal and the ecological conditions supporting actual populations.

Data and tools

- **Occurrences:** supplied coordinate-only extracts from Artportalen. Observation dates and survey effort are not included in these teaching files.
- **Climate:** CHELSA V2.1, 1981–2010 baseline; BIO1, BIO12 and BIO15, at approximately 2.5 arc-minutes. CHELSA lists these data under CC0.
- **Boundary:** Natural Earth, 1:10m country boundaries (public domain).
- **Modelling:** SDMtune, Maxnet implementation.